

Molecules

In most materials, atoms are joined in tiny groups called molecules. The shapes of molecules and the way they pack together can help explain how different materials behave.

Steaming ahead

Molecules are always jiggling around. When they get hot, they move farther and faster. When water heats up, the molecules may start moving so fast that they escape into the air as water vapor.

Steam appears when water vapor cools down and becomes liquid again. The steam from this train is made of millions of tiny liquid droplets.

Frozen solid

Cold molecules move slowly, allowing them to pack together more easily. When water freezes, the molecules line up in neat rows, forming ice crystals.

Snow may look like white powder, but if you look closely you can see thousands of tiny crystals as clear as glass.

Melt: As a solid heats up, its molecules move faster until they break free from each other and move separately, turning the solid into a liquid.

Solidify: As a liquid cools, its molecules lose energy and move more slowly. Eventually, they start sticking together, turning the liquid into a solid.

Solid

Liquid

If a liquid is poured into a jar or bottle, it takes the shape of its container and stays in place.